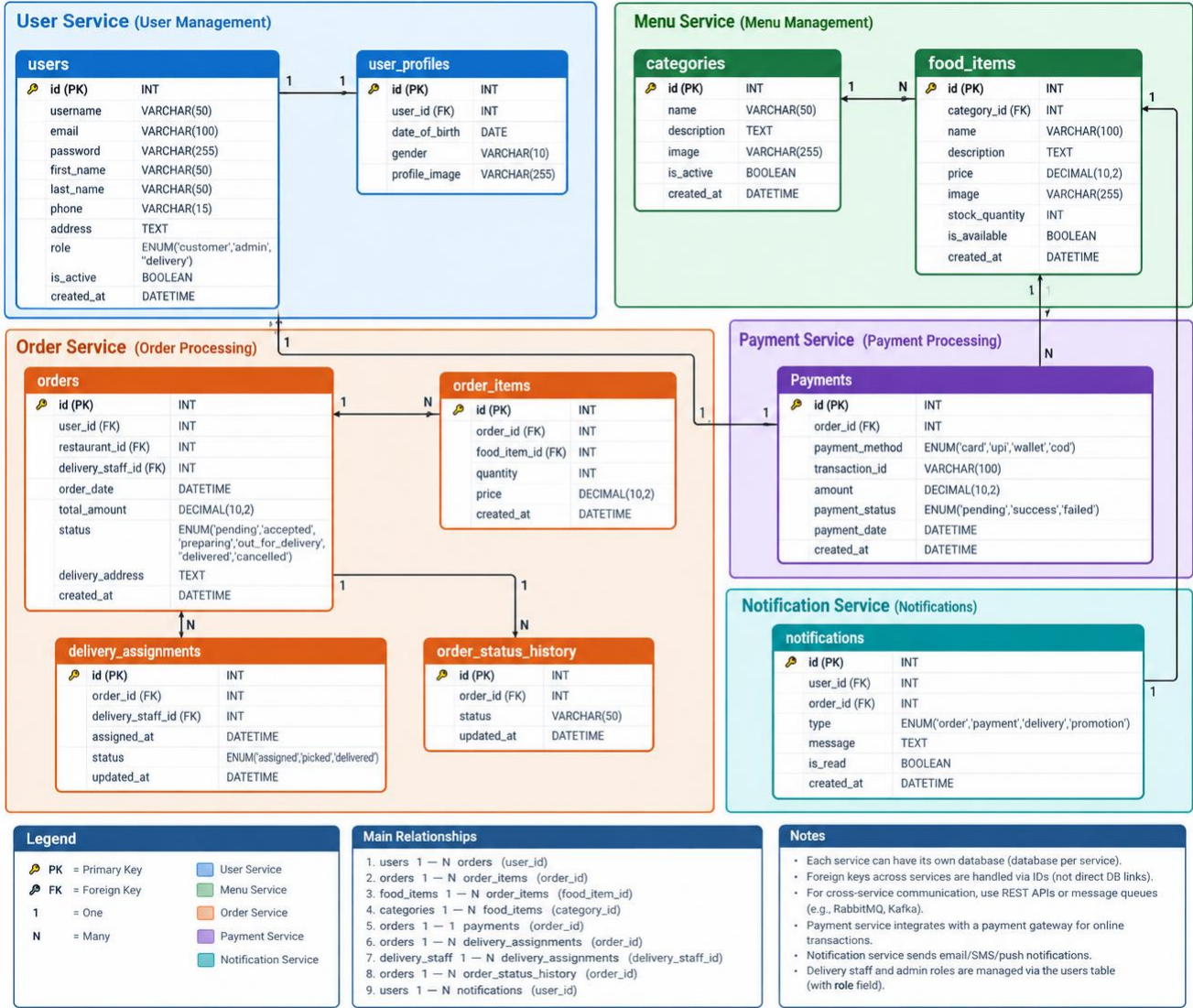


Project Schema

Distributed Microservices-Based Restaurant Website
Database ER Diagram (with Fields)



Complete project connection:-

restaurant_project/

|

|—— user_service/

|

|—— manage.py

|

|

|

|—— user_service/

|

|

|—— settings.py

|

|

|—— urls.py

|

|

|

|—— users/

|

|—— models.py

|

|—— serializers.py

|

|—— views.py

|

|—— urls.py

|

|—— admin.py

|

|—— menu_service/

|

|—— manage.py

|

|—— menu_service/

|

|

|—— settings.py

|

|

|—— urls.py

```

|   |
|   └─ menu/
|       └─ models.py
|       └─ serializers.py
|       └─ views.py
|       └─ urls.py
|
|   └─ order_service/
|       └─ orders/
|
|   └─ payment_service/
|       └─ payments/
|
|   └─ delivery_service/
|       └─ deliveries/
|
|   └─ notification_service/
|       └─ notifications/

```

Step:1

Browser ↓ Django Server ↓ project urls.py ↓ app urls.py ↓ views.py ↓ HTML response

Step:2

Login page:-

In the website:-

LOGIN PAGE APPEARS → Email,Password → LOGIN

IN INNER PROJECT

login.html→POST /login/ →users/urls.py→views.py→models.py→USERS
TABLE

LOGIN FOR FOOD ORDERING:-

LOGIN



AUTHENTICATION



CUSTOMER DASHBOARD



MENU



FOOD DETAILS



ADD TO CART



CART



CHECKOUT



ORDER



PAYMENT

↓
ORDER TRACKING

↓
DELIVERY

↓
NOTIFICATION

STEP:3

AFTER SUCESSFUL LOGIN IT WILL REDIRECT TO DASH BOARD

LOGIN --→ DASH BOARD

login_user()

↓
redirect()

↓
/dashboard/

↓
dashboard()

↓
dashboard.html

STEP:4

Dashboard → Menu

dashboard.html

↓
/menu/

↓

menu/urls.py



menu/views.py



FoodItem model



food_items table



menu.html

STEP:5

Menu → Database

views.py



FoodItem



food_items table



SELECT available food

STEP:5

Menu → Add to Cart

IF SUPPOSE THE CUSTOMER CLICK ADD TO CART

|

THE BROWSER SENDS TO (POST /cart/add/5/)

|

THE FLOW WILL BE

menu.html



POST /cart/add/5/



cart/urls.py



add_to_cart()



FoodItem



Cart



CartItem



DATABASE

STEP:6

Cart → Checkout

CUSTOMER CLICKS CHECK OUT

|

THE FLOW WIL BE AS:

cart.html



/checkout/



checkout()



Cart



CartItem



Customer Address



Checkout Page

➔ THIS PAGE COLLECTS ➔ Name,Phone,Delivery Address,Pincode

Step-7

Checkout → Order

When the customer press place order the code creates orders

+

order_items

The Flow of the process is:

CHECKOUT



POST /orders/create/



Order Service



Order View



Order Model



ORDERS TABLE



ORDER_ITEMS TABLE

STEP8:

Order → Payment

After creating the order: return redirect to payments

ORDER CREATED



PAYMENT URL



Payment View



Payment Gateway



PAYMENTS TABLE

STEP9:

Payment → Order Status

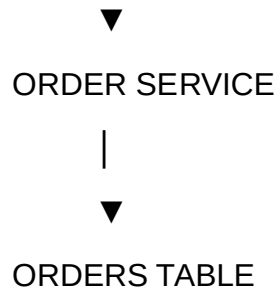
Suppose payment succeeds. The payment service records: payment_status= Success

Then the order service can update: order_status = Accepted

PAYMENT SERVICE



| REST API



STEP9:

Order → Delivery

After restaurant preparation:

Pending



Accepted



Preparing



Ready



Delivery Staff Assigned



Out for Delivery



Delivered

Programming flow:

Order Service



Delivery Service



deliveries table

The delivery staff can then update:

delivery_status = "Out for Delivery" and finally: delivery_status = "Delivered"

STEP10:

Order → Notification

Whenever an important event happens:

Order Created



Notification Service



notifications table



Customer

For example:

"Your order #1001 has been confirmed."

Your document specifically lists order confirmation, payment confirmation, order status and delivery updates as notification functions.